Summary 1

Systems of Linear Equations

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1.1 Systems of Linear Equations

1.1.1 Forming the Equations

Solution		Comments
	$a+b+c=176$ $0.85a+0.90b+0.95c=161.80$ $0.93a+0.95b+0.98c=169.56$ Using GC, $a=32$ $b=44$ $c=100$	

1.1.2 GC

Two linear equations are independent if and only if neither equation is a linear combination of the other.

Solution		Comments
	$x+y+z=6$ $2x-y+z=3$ $x+2y-z=2$ Using GC, $x=1$ $y=2$ $z=3$	

1.1.3 Special Case: Number of Equations > Number of Variables

Sometimes there may be no solution, e.g. skew lines in vectors.

1.1.4 Special Case: Number of Equations < Number of Variables

Solution		Comments
	$x+y+z=6$ $x-y+2z=5$ Using GC, $x=\frac{11}{2}-\frac{3}{2}z$ $y=\frac{1}{2}+\frac{1}{2}z$ $z=z$	

Summary 2

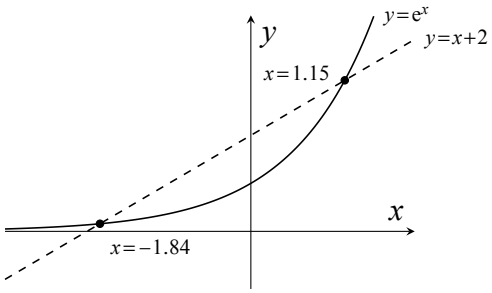
Inequalities

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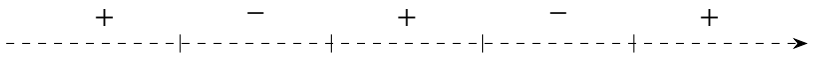
2.1 Solving Inequalities Graphically

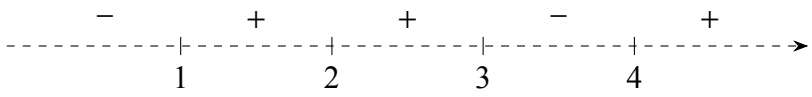
- Rearrange to $\mathrm{f}(x) > 0$, or compare $y = \mathrm{f}(x)$ with $y = \mathrm{g}(x)$.
- Sketch on the GC; mark the intersections and the asymptotes.
- Read off the x-values where the required graph is above or below.

Solution		Comments
	$\mathrm{e}^x < x + 2$  Using GC, intersections at $x = -1.8414$, $x = 1.1462$ $\therefore\ -1.84 < x < 1.15$	

2.2 Sign Test Technique

- Make the right-hand side 0.
- Do **not** cross multiply unless the expression or bracket is always positive.
- Expressions that are always positive can be divided across, e.g. a quadratic with discriminant <0 and positive leading coefficient: $x^2+2x+5=(x+1)^2+4>0$.



Solution	Comments
$\dfrac{(x-1)(x-2)^2}{(x-3)(x-4)} \geqslant 0$  $x=2$: numerator $=0$ (included) $x=3,\ x=4$: denominator $=0$ (excluded) $\therefore 1 \leqslant x<3$ or $x>4$	

Note

Crossing a factor with an odd power changes the sign; crossing a factor with an even power keeps the same.

2.3 Common Substitutions

2.3.1 General Example

Solution		Comments
	$x^2-x-2<0$ $(x+1)(x-2)<0$ $-1<x<2$ $x\to\lvert x\rvert:\quad -1<\lvert x\rvert<2$ $\lvert x\rvert<2 \quad (\because \lvert x\rvert\geqslant 0>-1)$ $-2<x<2$ $x\to\frac{1}{x}:\quad -1<\frac{1}{x}<2$ $x<-1$ or $x>\frac{1}{2}$ (from the graph of $y=\frac{1}{x}$)	

2.3.2 Substitution Tables

Base results: (A) $x>2$ (B) $x<-1$ (C) $-1<x<2$.

$x\to e^x$

(A)	(B)	(C)
$e^x>2$	$e^x<-1$	$-1<e^x<2$
$x>\ln 2$	no solution	$e^x<2\ (\because e^x>0)$
	$(\because e^x>0)$	$x<\ln 2$

$x\to \ln x\ (x>0)$

(A)	(B)	(C)
$\ln x>2$	$\ln x<-1$	$-1<\ln x<2$
$x>e^2$	$0<x<e^{-1}$	$e^{-1}<x<e^2$

$x\to -x$

(A)	(B)	(C)
$-x>2$	$-x<-1$	$-1<-x<2$
$x<-2$	$x>1$	$-2<x<1$

$x \to x+1$

(A)	**(B)**	**(C)**
$x+1>2$	$x+1<-1$	$-1<x+1<2$
$x>1$	$x<-2$	$-2<x<1$

$x \to \lvert x\rvert$

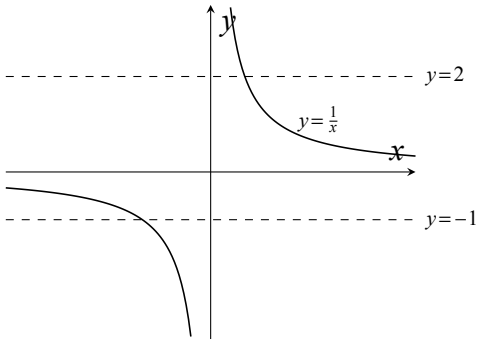
(A)	**(B)**	**(C)**
$\lvert x\rvert>2$	$\lvert x\rvert<-1$	$-1<\lvert x\rvert<2$
$x<-2$ or $x>2$	no solution	$\lvert x\rvert<2$
	$(\because \lvert x\rvert \geqslant 0)$	$-2<x<2$

$x \to x^2$

(A)	**(B)**	**(C)**
$x^2>2$	$x^2<-1$	$-1<x^2<2$
$x<-\sqrt{2}$ or $x>\sqrt{2}$	no solution	$x^2<2$
	$(\because x^2 \geqslant 0)$	$-\sqrt{2}<x<\sqrt{2}$

$x \to 1/x$ (VERY IMPORTANT: draw the graph)

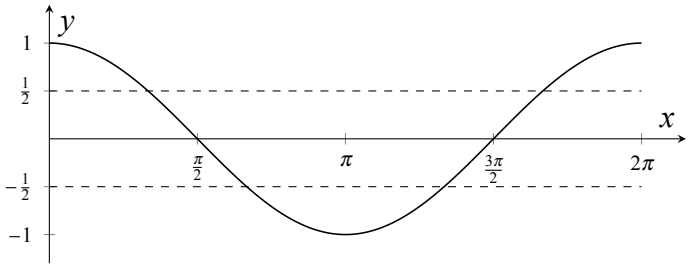
(A)	**(B)**	**(C)**
$\frac{1}{x}>2$	$\frac{1}{x}<-1$	$-1<\frac{1}{x}<2$



Since $-1 \leqslant \cos x \leqslant 1$, the cases are adapted to (A) $x>\frac{1}{2}$, (B) $x<-\frac{1}{2}$, (C) $-\frac{1}{2}<x<\frac{1}{2}$.

$x \to \cos x$ for $0 \le x \le 2\pi$ (draw the graph)

(A)	**(B)**	**(C)**
$\cos x > \frac{1}{2}$	$\cos x < -\frac{1}{2}$	$-\frac{1}{2} < \cos x < \frac{1}{2}$



Solution	**Comments**
$$\frac{1}{2} < \cos\theta < \frac{\sqrt{2}}{2}, \quad 0 \leqslant \theta \leqslant 3\pi$$ 